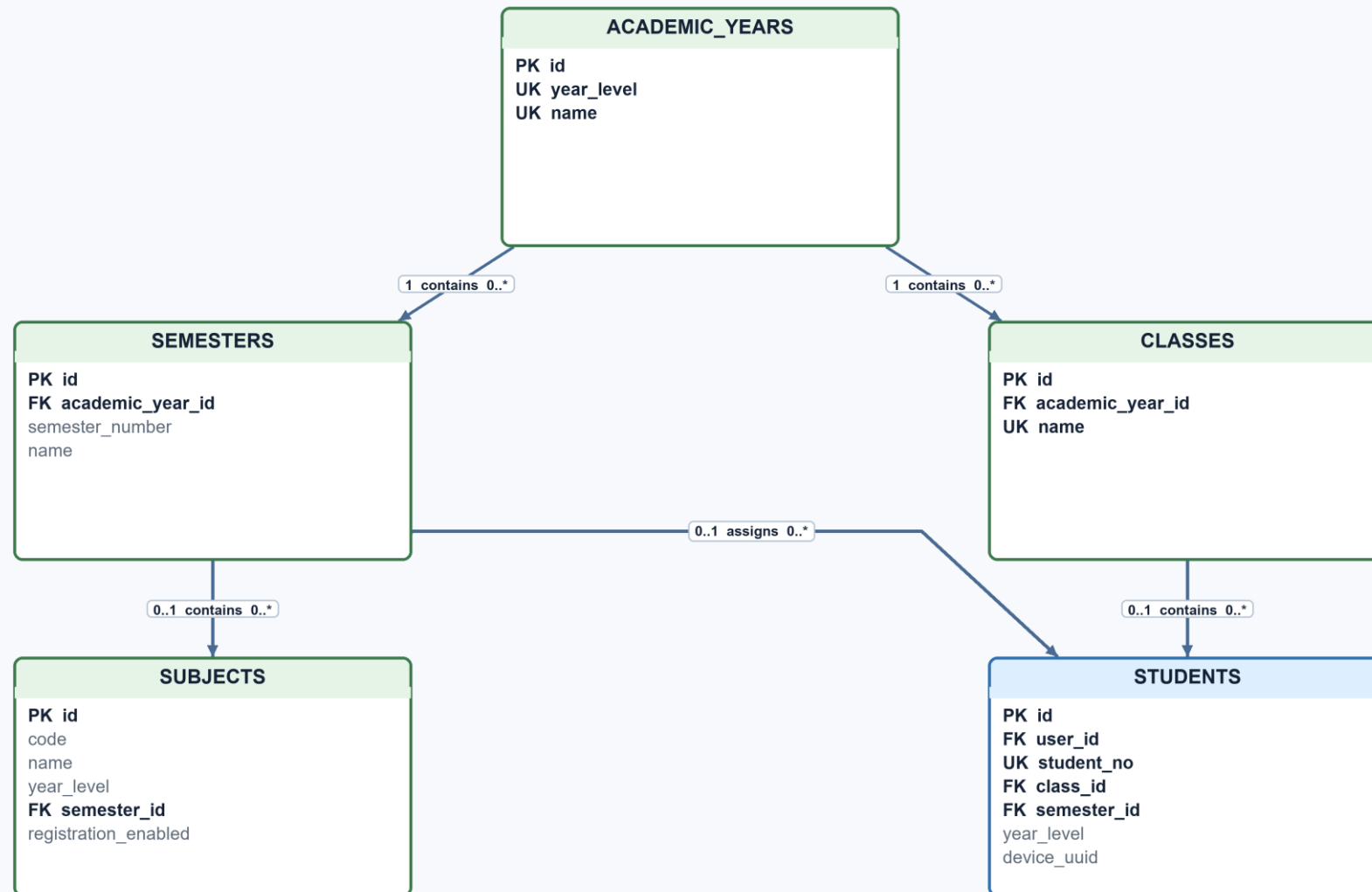


EasyAttend Academic Structure Relationships

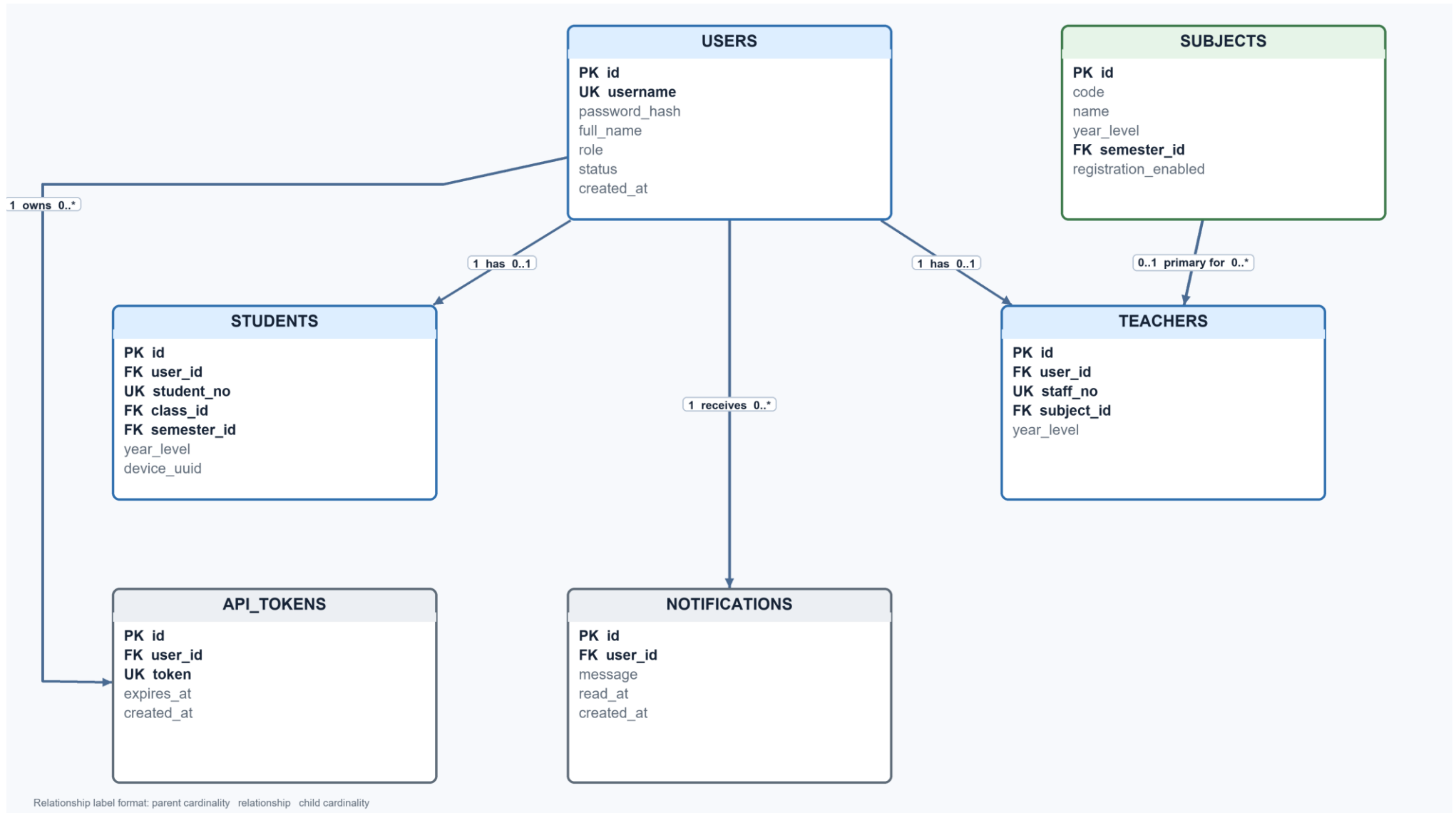
Academic years organize semesters and classes. Semesters organize subjects, while students are assigned to a semester and class.



Relationship label format: parent cardinality relationship child cardinality

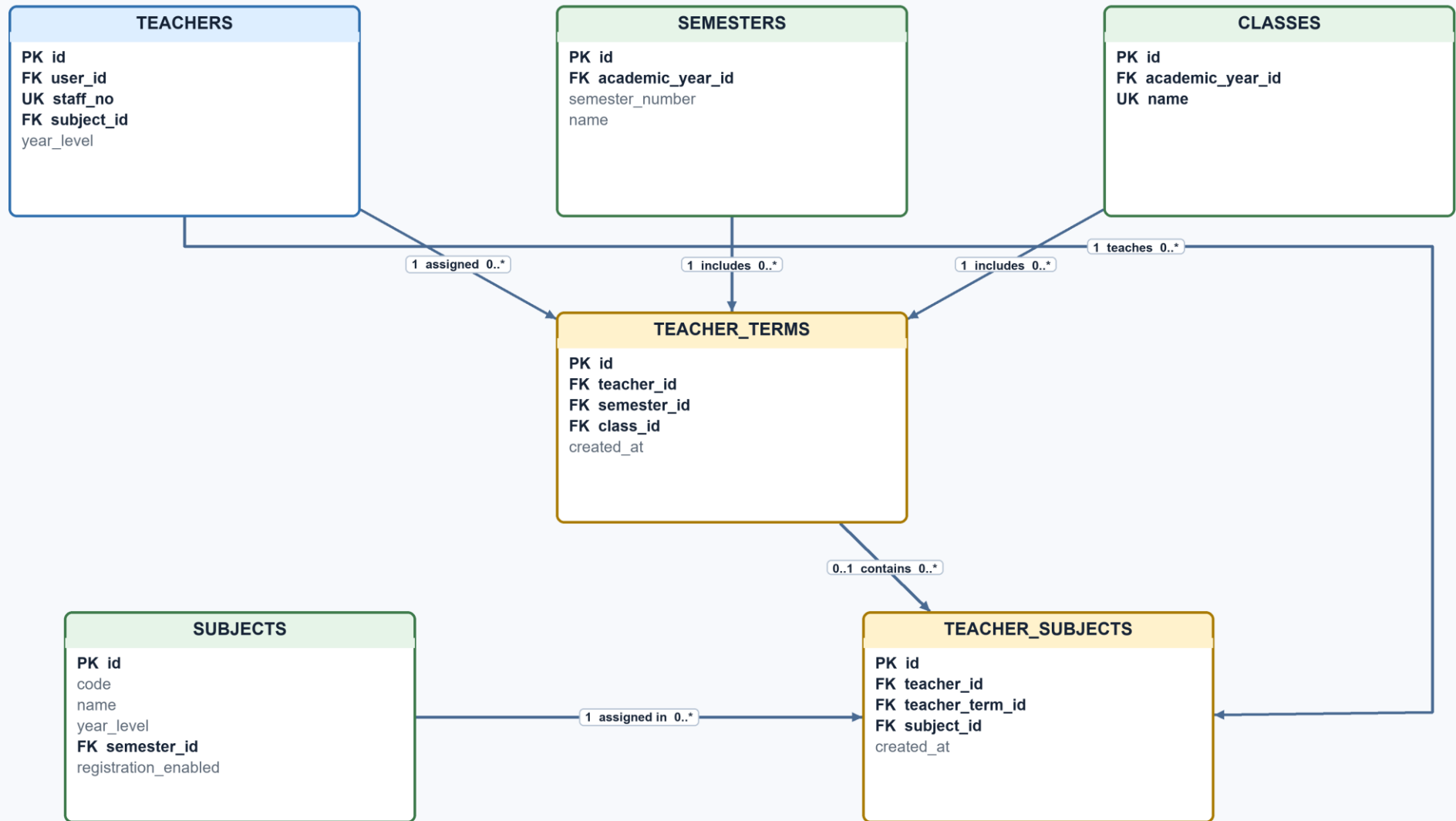
User Profile and Support Relationships

Users provide the shared login identity for student and teacher profiles. The same user record owns authentication tokens and receives notifications.



Teaching Assignment Relationships

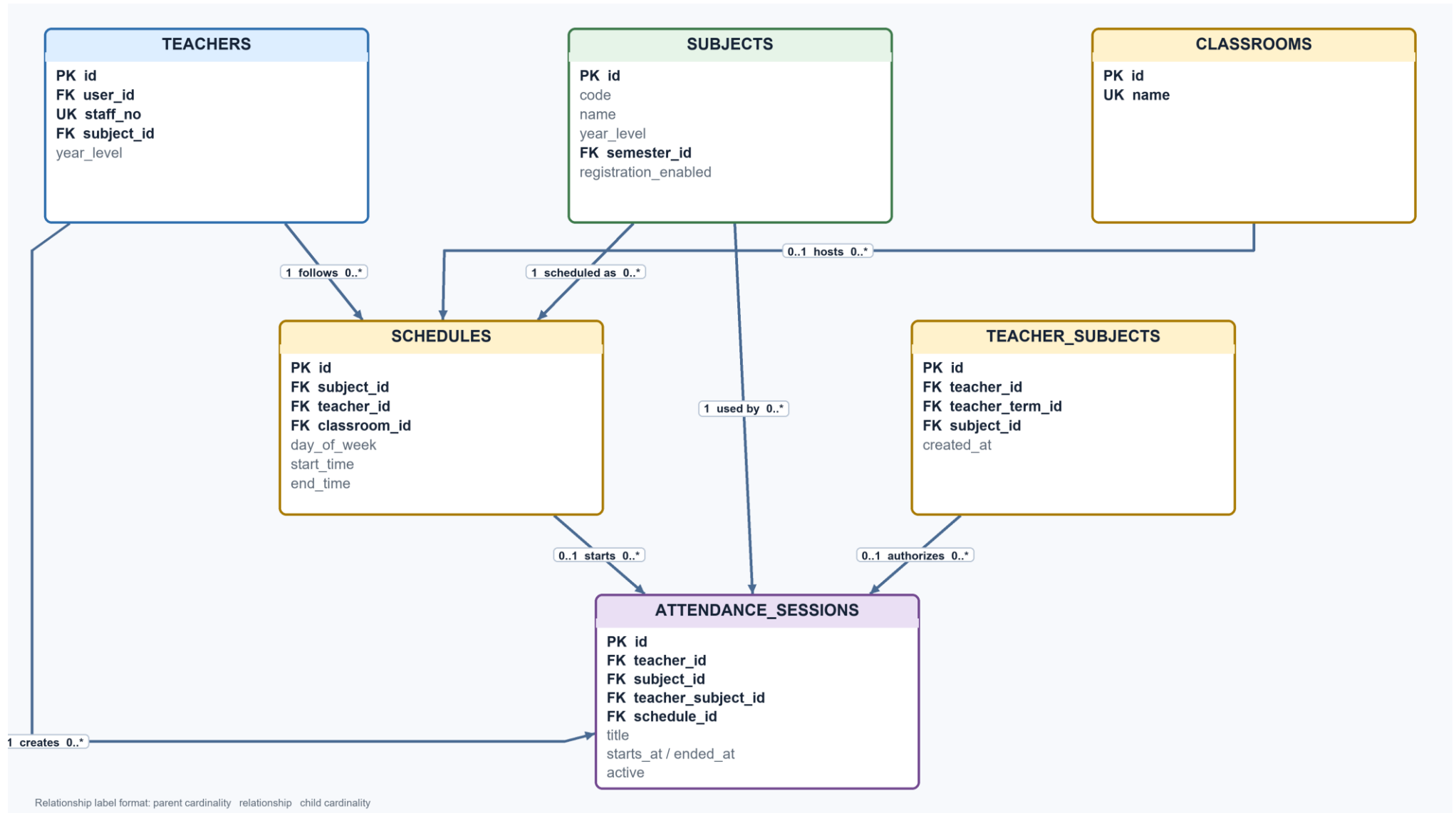
Teacher term records connect a teacher to a semester and class. Teacher subject records then connect that term assignment to the subjects being taught.



Relationship label format: parent cardinality relationship child cardinality

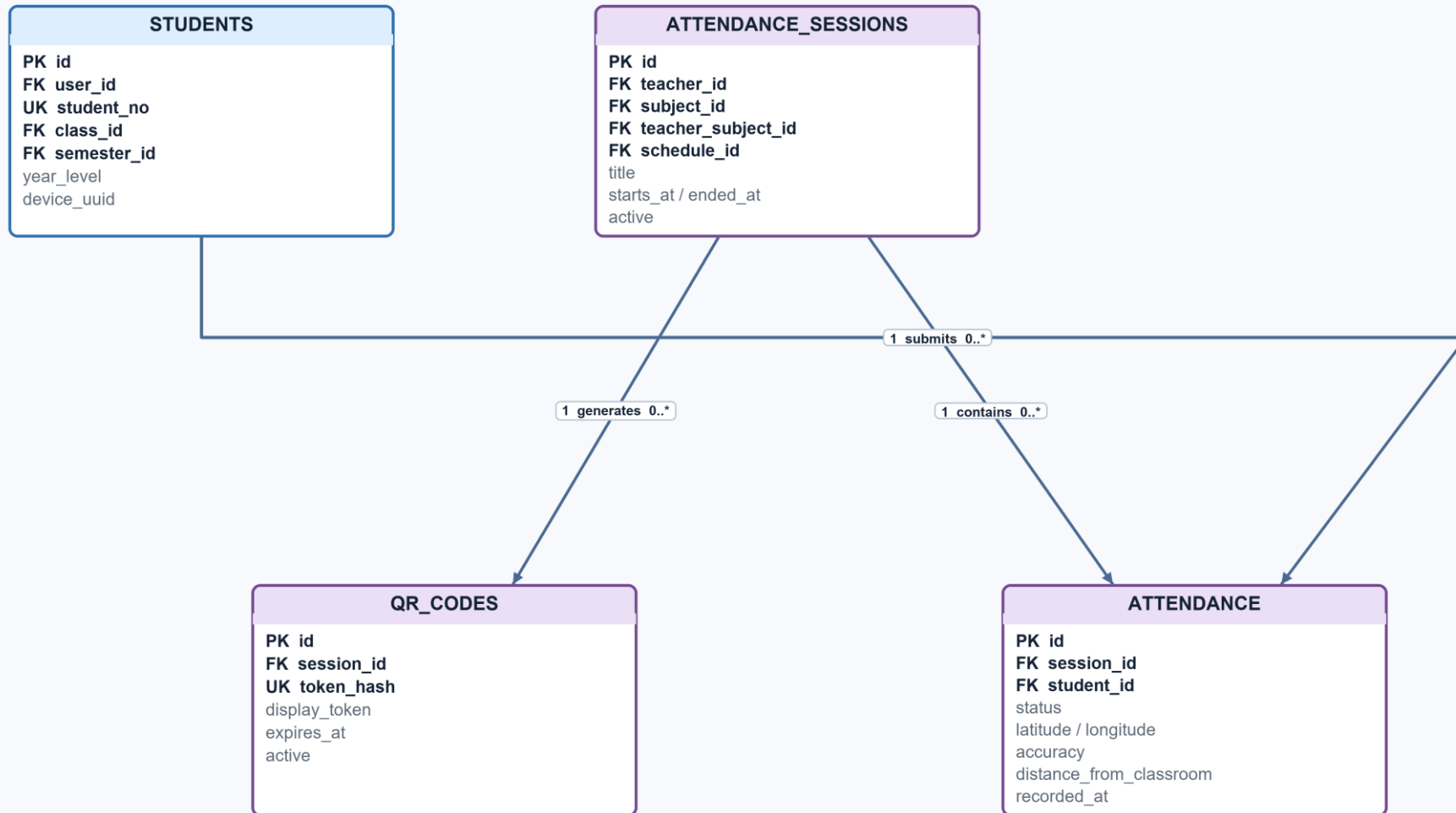
Scheduling and Attendance Session Relationships

Schedules connect teachers, subjects, and classrooms. Attendance sessions are created by teachers and may reference a schedule and an authorized teacher subject assignment.



QR GPS and Student Attendance Relationships

An attendance session generates QR codes and attendance records. Each attendance record belongs to one student and stores GPS coordinates, accuracy, and distance from the configured classroom area.



Complete Database Relationship Reference

This reference lists every foreign key relationship represented across the five diagrams. Cardinality is shown from the parent entity to the child entity.

Parent entity	Cardinality	Child entity	Foreign key	Relationship meaning
USERS	1 to 0..1	STUDENTS	students.user_id	A user may have one student profile
USERS	1 to 0..1	TEACHERS	teachers.user_id	A user may have one teacher profile
USERS	1 to 0..*	NOTIFICATIONS	notifications.user_id	A user receives notifications
USERS	1 to 0..*	API_TOKENS	api_tokens.user_id	A user owns API login tokens
ACADEMIC_YEARS	1 to 0..*	SEMESTERS	semesters.academic_year_id	An academic year contains semesters
ACADEMIC_YEARS	1 to 0..*	CLASSES	classes.academic_year_id	An academic year contains classes
SEMESTERS	0..1 to 0..*	SUBJECTS	subjects.semester_id	A semester contains subjects
SEMESTERS	0..1 to 0..*	STUDENTS	students.semester_id	Students are assigned to a semester
CLASSES	0..1 to 0..*	STUDENTS	students.class_id	A class contains students
SUBJECTS	0..1 to 0..*	TEACHERS	teachers.subject_id	A teacher may retain a primary subject
TEACHERS	1 to 0..*	TEACHER_TERMS	teacher_terms.teacher_id	A teacher receives term assignments
SEMESTERS	1 to 0..*	TEACHER_TERMS	teacher_terms.semester_id	A term assignment belongs to a semester
CLASSES	1 to 0..*	TEACHER_TERMS	teacher_terms.class_id	A term assignment belongs to a class
TEACHERS	1 to 0..*	TEACHER_SUBJECTS	teacher_subjects.teacher_id	A teacher teaches assigned subjects
TEACHER_TERMS	0..1 to 0..*	TEACHER_SUBJECTS	teacher_subjects.teacher_term_id	A term assignment contains subject assignments
SUBJECTS	1 to 0..*	TEACHER_SUBJECTS	teacher_subjects.subject_id	A subject appears in teaching assignments
SUBJECTS	1 to 0..*	SCHEDULES	schedules.subject_id	A subject has scheduled periods
TEACHERS	1 to 0..*	SCHEDULES	schedules.teacher_id	A teacher follows scheduled periods
CLASSROOMS	0..1 to 0..*	SCHEDULES	schedules.classroom_id	A classroom hosts scheduled periods
TEACHERS	1 to 0..*	ATTENDANCE_SESSIONS	attendance_sessions.teacher_id	A teacher creates attendance sessions
SUBJECTS	1 to 0..*	ATTENDANCE_SESSIONS	attendance_sessions.subject_id	A session records attendance for a subject

Parent entity	Cardinality	Child entity	Foreign key	Relationship meaning
TEACHER_SUBJECTS	0..1 to 0..*	ATTENDANCE_SESSIONS	attendance_sessions.teacher_subject_id	An assignment authorizes a session
SCHEDULES	0..1 to 0..*	ATTENDANCE_SESSIONS	attendance_sessions.schedule_id	A schedule may start attendance sessions
ATTENDANCE_SESSIONS	1 to 0..*	QR_CODES	qr_codes.session_id	A session generates QR codes
ATTENDANCE_SESSIONS	1 to 0..*	ATTENDANCE	attendance.session_id	A session contains attendance records
STUDENTS	1 to 0..*	ATTENDANCE	attendance.student_id	A student submits attendance records